Theorem 2.3, (b) ⇒ (c): extreme point ⇒ basic feasible solution

Based on Theorem 2.3 (Section 2.2) of Bertsimas and Tsitsiklis [1]

Source. This note expands the proof of (b) ⇒ (c) in Theorem 2.3 of [1, Section 2.2]. Definition 2.9, Theorem 2.2 and Figure 2.6 below refer to the same book.

Setting. Let $P \subset \mathbb{R}^n$ be a polyhedron described by finitely many constraints

$$\mathbf{a}_i'\mathbf{x} \geq b_i \quad (i \in M_1), \qquad \mathbf{a}_i'\mathbf{x} = b_i \quad (i \in M_2),$$

where a constraint $\leq$ is rewritten as $\geq$ by multiplying by -1. For $\mathbf{x}^* \in P$ let

$$I = \{\, i : \mathbf{a}_i'\mathbf{x}^* = b_i \,\}$$

be the set of constraints active at $\mathbf{x}^*$. Note $M_2 \subseteq I$ because $\mathbf{x}^*$ is feasible.

Claim. *If $\mathbf{x}^* \in P$ is an extreme point of P, then $\mathbf{x}^*$ is a basic feasible solution.*

Strategy: contrapositive

We prove the equivalent statement: *if $\mathbf{x}^* \in P$ is not a basic feasible solution, then $\mathbf{x}^*$ is not an extreme point.*

To show $\mathbf{x}^*$ is not extreme, by definition we must exhibit $\mathbf{y}, \mathbf{z} \in P$, both different from $\mathbf{x}^*$, and $\lambda \in [0,1]$ with $\mathbf{x}^* = \lambda\mathbf{y} + (1-\lambda)\mathbf{z}$. We will build $\mathbf{y} = \mathbf{x}^* + \epsilon\mathbf{d}$ and $\mathbf{z} = \mathbf{x}^* - \epsilon\mathbf{d}$, so that $\mathbf{x}^*$ is their midpoint ($\lambda = 1/2$). The whole proof is about finding a direction $\mathbf{d}$ and a step ϵ that keep both points inside P.

Step 1: what "not basic feasible" means

Since $\mathbf{x}^* \in P$, it satisfies every constraint, so condition (i) of Definition 2.9 in [1] (all equality constraints active) holds automatically, and $\mathbf{x}^*$ is feasible. Hence the only way $\mathbf{x}^*$ can fail to be a basic feasible solution is that condition (ii) fails:

$$\text{among } \{\mathbf{a}_i : i \in I\} \text{ there are fewer than } n \text{ linearly independent vectors.}$$

By Theorem 2.2 of [1], this is equivalent to

$$S := \operatorname{span}\{\mathbf{a}_i : i \in I\} \;\subsetneq\; \mathbb{R}^n.$$

Step 2: a direction that does not move the active constraints

Since S is a proper subspace, $\dim S < n$, hence $\dim S^\perp = n - \dim S \geq 1$ and there is $\mathbf{d} \neq \mathbf{0}$ with $\mathbf{d} \in S^\perp$, i.e.

$$\mathbf{a}_i'\mathbf{d} = 0 \qquad \text{for all } i \in I.$$

Equivalently: the matrix A_I whose rows are $\mathbf{a}_i'$, $i \in I$, has rank $< n$, so its null space is nontrivial.

Interpretation: moving from $\mathbf{x}^*$ along $\pm\mathbf{d}$ does not change the value of any active constraint. Indeed, for every $\epsilon \in \mathbb{R}$ and $i \in I$,

$$\mathbf{a}_i'(\mathbf{x}^* \pm \epsilon\mathbf{d}) = \mathbf{a}_i'\mathbf{x}^* \pm \epsilon\,\mathbf{a}_i'\mathbf{d} = b_i.$$

So every active constraint (in particular every equality constraint) remains satisfied, with equality, at both $\mathbf{y}$ and $\mathbf{z}$, *no matter how large* ϵ is.

Step 3: the inactive constraints, and why ϵ must be small

For $i \notin I$ the constraint is an inequality with strict slack

$$s_i := \mathbf{a}_i'\mathbf{x}^* - b_i > 0.$$

Moving along $\pm\mathbf{d}$ changes the left-hand side by $\pm\epsilon\,\mathbf{a}_i'\mathbf{d}$, which could be negative, so we must not move too far. For any $\epsilon > 0$,

$$\mathbf{a}_i'(\mathbf{x}^* \pm \epsilon\mathbf{d}) = \mathbf{a}_i'\mathbf{x}^* \pm \epsilon\,\mathbf{a}_i'\mathbf{d} \;\geq\; \mathbf{a}_i'\mathbf{x}^* - \epsilon\,|\mathbf{a}_i'\mathbf{d}|.$$

Therefore, if

$$\epsilon\,|\mathbf{a}_i'\mathbf{d}| < \mathbf{a}_i'\mathbf{x}^* - b_i = s_i \qquad \text{for all } i \notin I, \tag{1}$$

then $\mathbf{a}_i'(\mathbf{x}^* \pm \epsilon\mathbf{d}) > b_i$ for all $i \notin I$.

Such an $\epsilon > 0$ exists. If $\mathbf{a}_i'\mathbf{d} = 0$, condition (1) holds for every $\epsilon > 0$ since $s_i > 0$. Let $J = \{i \notin I : \mathbf{a}_i'\mathbf{d} \neq 0\}$. If $J = \emptyset$, any $\epsilon > 0$ works. Otherwise J is *finite* (a polyhedron has finitely many constraints), so

$$\epsilon := \tfrac{1}{2} \min_{i \in J} \frac{s_i}{|\mathbf{a}_i'\mathbf{d}|} > 0$$

is a minimum of finitely many positive numbers, hence positive, and it satisfies (1). Finiteness is essential here: with infinitely many constraints the infimum of the ratios could be 0.

Step 4: conclusion

With this ϵ, set $\mathbf{y} = \mathbf{x}^* + \epsilon\mathbf{d}$ and $\mathbf{z} = \mathbf{x}^* - \epsilon\mathbf{d}$.

- By Step 2, $\mathbf{y}$ and $\mathbf{z}$ satisfy every constraint $i \in I$ (with equality).
- By Step 3, $\mathbf{y}$ and $\mathbf{z}$ satisfy every constraint $i \notin I$ (strictly).

Hence $\mathbf{y}, \mathbf{z} \in P$. Since $\epsilon > 0$ and $\mathbf{d} \neq \mathbf{0}$, we have $\mathbf{y} \neq \mathbf{x}^*$ and $\mathbf{z} \neq \mathbf{x}^*$. Finally

$$\tfrac{1}{2}\mathbf{y} + \tfrac{1}{2}\mathbf{z} = \tfrac{1}{2}(\mathbf{x}^* + \epsilon\mathbf{d}) + \tfrac{1}{2}(\mathbf{x}^* - \epsilon\mathbf{d}) = \mathbf{x}^*,$$

so $\mathbf{x}^*$ is a convex combination (with $\lambda = 1/2$) of two points of P different from it. Therefore $\mathbf{x}^*$ is not an extreme point. This proves the contrapositive, and hence (b) $\Rightarrow$ (c). $\square$

The proof on the example of Figure 2.6 of [1]

Example. $P = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1,\ x_1, x_2, x_3 \geq 0\}$, and $E = (x_1^*, 0, x_3^*)$ with $x_1^*, x_3^* > 0$.

Step 1. Active set: $I = \{x_1 + x_2 + x_3 = 1,\ x_2 \geq 0\}$ with vectors $(1, 1, 1)$ and $(0, 1, 0)$. Only $2 < 3$ independent vectors, so E is not basic and S is a plane in $\mathbb{R}^3$.

Step 2. $S^\perp$ is the line spanned by $\mathbf{d} = (1, 0, -1)$: indeed $(1, 1, 1) \cdot \mathbf{d} = 0$ and $(0, 1, 0) \cdot \mathbf{d} = 0$. Moving along $\pm\mathbf{d}$ keeps $x_1 + x_2 + x_3 = 1$ and $x_2 = 0$; geometrically, we slide along the edge AB.

Step 3. Inactive constraints: $x_1 \geq 0$ with $\mathbf{a} = (1, 0, 0)$, slack $s = x_1^*$, $|\mathbf{a}'\mathbf{d}| = 1$; and $x_3 \geq 0$ with $\mathbf{a} = (0, 0, 1)$, slack $s = x_3^*$, $|\mathbf{a}'\mathbf{d}| = 1$. So we need $\epsilon < \min\{x_1^*, x_3^*\}$: this is exactly the condition for not crossing past B (where x_3 would become negative) or past A (where x_1 would become negative).

Step 4. $\mathbf{y} = (x_1^* + \epsilon,\ 0,\ x_3^* - \epsilon)$ and $\mathbf{z} = (x_1^* - \epsilon,\ 0,\ x_3^* + \epsilon)$ lie on the segment AB, inside P, and E is their midpoint. So E is not extreme.

Example (Why the argument fails at a basic feasible solution). At $A = (0, 0, 1)$, I contains $(1, 1, 1)$, $(1, 0, 0)$, $(0, 1, 0)$, which span $\mathbb{R}^3$. Then $S^\perp = \{\mathbf{0}\}$: the only $\mathbf{d}$ with $\mathbf{a}_i'\mathbf{d} = 0$ for all $i \in I$ is $\mathbf{d} = \mathbf{0}$, so Step 2 cannot produce a direction and $\mathbf{y} = \mathbf{z} = A$. Any nonzero direction changes some active constraint, so one of $A \pm \epsilon\mathbf{d}$ violates it (e.g. $\mathbf{d} = (1, -1, 0)$ keeps the equality but $A - \epsilon\mathbf{d}$ has $x_1 = -\epsilon < 0$).

Remark (Summary of the idea). Not basic $\Rightarrow$ the active constraints leave a free direction $\mathbf{d}$ (Step 2). Along $\pm\mathbf{d}$ the active constraints never break, and the inactive ones have positive slack, so a small enough step keeps both $\mathbf{x}^* \pm \epsilon\mathbf{d}$ feasible (Step 3). Then $\mathbf{x}^*$ is the midpoint of a segment in P, i.e. not extreme.

References

[1] D. Bertsimas and J. N. Tsitsiklis, *Introduction to Linear Optimization*, Athena Scientific, Belmont, MA, 1997.